Aryan Ballani

Data Engineering, Analytics, and Scalable Pipelines

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Education

BSc. Computer Science and Statistics | University of British Columbia *Sep 2021 – Apr 2026*

- **Dean's Honor List** (sessional standing for 3 consecutive years, GPA: 4.0/4.33).
- Awarded the **International Student Scholarship** for community presence and academic excellence.
- **Hackathon Mentor**, BCS Hacks 2025: Mentored 100+ participants on project ideation, implementation, and debugging.

Work Experience

Machine Learning Engineer — *Railtown AI Inc., Vancouver, BC* *Jan 2025 – Present*

- Built and maintained **data ingestion pipelines** processing metadata from **10K+ commits**, automating issue triaging and enabling Root Cause Analysis and Commit Summarizer tools.
- Developed and open-sourced **Railtracks framework** using **pub-sub architecture** and **microservices**: adopted by 20+ developers, reducing LLM integration complexity by **40%**.
- Engineered **20+ RESTful API endpoints** following modular design patterns, achieving scalable async execution across multi-model backends.
- Led full **test infrastructure overhaul** (5,100+ lines across 125 files), lifting coverage to **91%** and cutting unit-test runtime **10×** (76s to 7.6s); suite now exceeds **2,200 tests**.
- Dockerized and deployed a reproducible **benchmarking pipeline** using SWE-bench, supporting standardized evaluation across 3+ internal models.
- Packaged and deployed the Railtracks library to **Azure Artifacts** with GitHub Actions, streamlining distribution for 5 microservices.

Software Engineer in Test — *BC Liquor Distribution Branch, Burnaby, BC* *May 2024 – Dec 2024*

- Automated **80% of manual test cases** using **Selenium** and **Oracle SQL**, raising QA throughput by **25%**.
- Enhanced application performance by **12%** through profiling and optimizing **30+ database queries** and integrating **100+ unit tests**.
- Spearheaded **CI/CD integration** with Azure DevOps, reducing release cycle time by **30%** across 5 QA modules.

Undergraduate Teaching Assistant — *UBC Computer Science Dept., Vancouver, BC* *Jan 2024 – Apr 2026*

- Led labs and practicals for **50+** students weekly, supporting hands-on learning and clarity in technical explanations.
- Answered **200+** student queries per term and received praise for engagement and instructional communication.

Relevant Projects

OrbitX Data — Python, XGBoost, GeoPandas, NASA APIs *Apr 2025*

- **Multi-Source Data Curation:** Automated ingestion from NASA POWER API (climate variables) and NASA FIRMS (fire detections), producing **60,000+ labeled training records**.
- **Threshold-Tuned Classifier:** Trained XGBoost model, lifting wildfire recall to **75%** at 0.6 threshold while holding accuracy above **92%**.
- **Datathon-Winning Delivery:** Placed **1st overall** at OrbitX Datathon 2025 among competing student teams.

NYC Airbnb Regression Analysis — R, tidyverse, ggplot2 *Jan 2024*

- **EDA and Feature Engineering:** Analyzed **5,000+ NYC Airbnb listings**, normalizing distributions and deriving new features to enhance regression model reliability.
- **Model Building and Validation:** Trained multiple models with Mallow's Cp and AIC; selected finalist ($R^2 = 0.676$, RMSE = 2.64) and validated with cross-validation and residual diagnostics.

Technical Skills

Languages	: Python, SQL, R, TypeScript, Java
Databases / Data Systems	: PostgreSQL, Oracle SQL, MongoDB, Apache Spark, pgvector
Data Tools	: Pandas, NumPy, Scikit-learn, XGBoost, NLTK, GeoPandas
Cloud / DevOps	: AWS, Azure, Docker, GitHub Actions, Azure DevOps
Concepts	: Data Pipelines, ETL, Data Quality, REST APIs, Distributed Systems, Linux